

pybounds Diagnostic Report — Mono-Camera System

States

g — ground speed

d — distance above ground

Dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, u)$

$$\dot{g} = u$$

$$\dot{d} = 0$$

Measurement $\mathbf{y} = h(\mathbf{x})$

$$r = g/d \quad (\text{ventral optic flow})$$

Simulation parameters

$$\mathbf{x}_0 = [g_0, d_0] = [2.0, 3.0]$$

$$dt = 0.01 \text{ s}$$

$$u(t) = \sin(3t) \text{ (with near-zero segment } 3 \leq t < 6 \text{ s)}$$

Observability parameters

Window size $w = 6$ time-steps

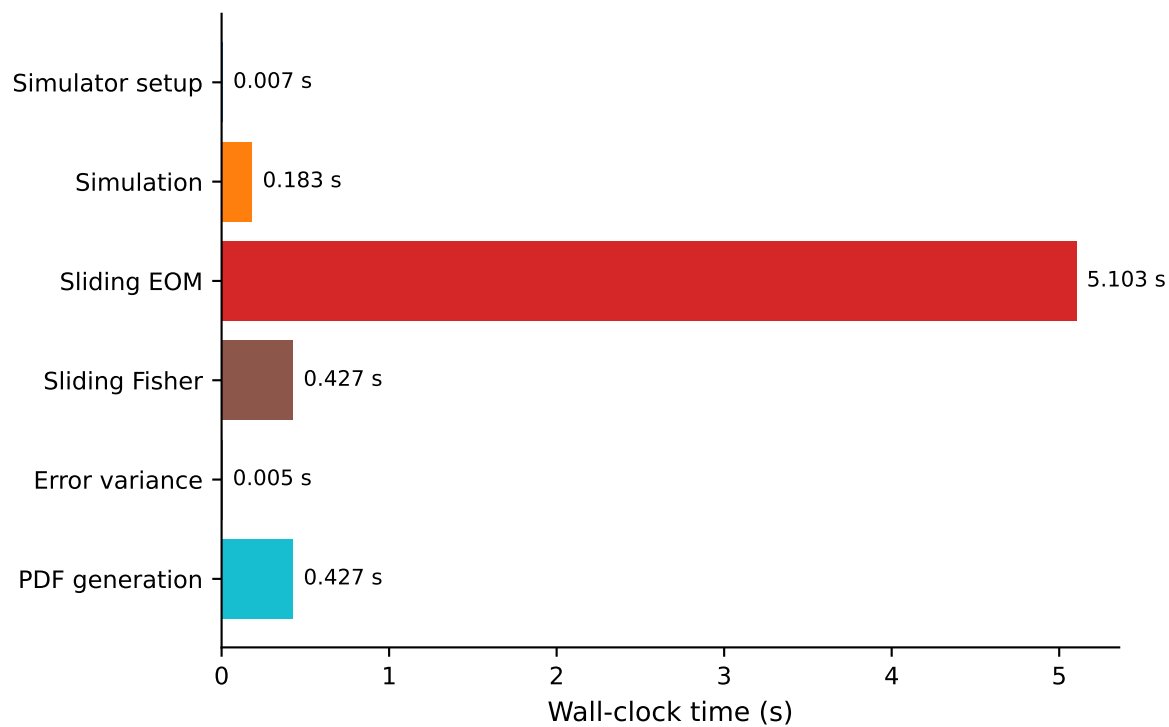
Perturbation $\varepsilon = 10^{-4}$

Sensor noise $R = \{r : 0.1\}$

Regularisation $\lambda = 10^{-8}$

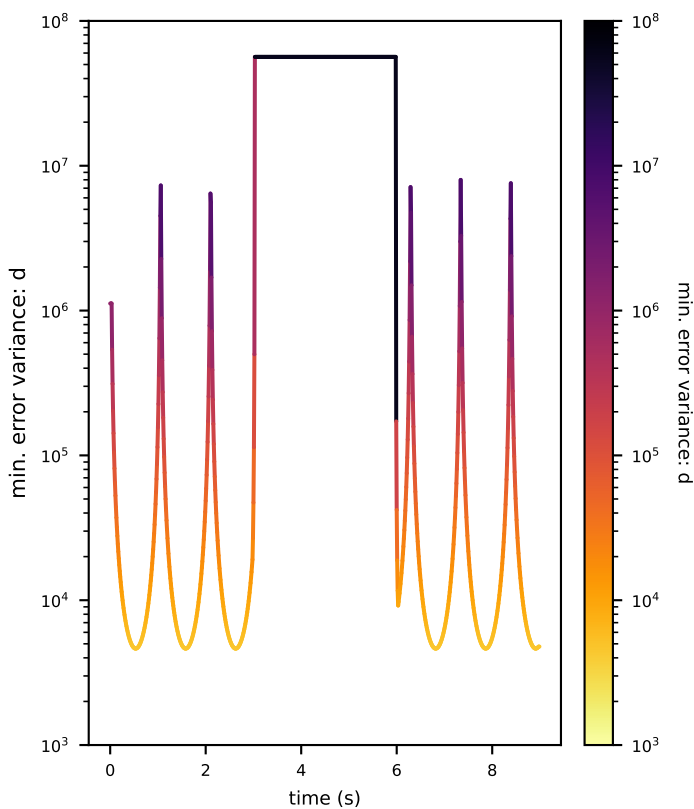
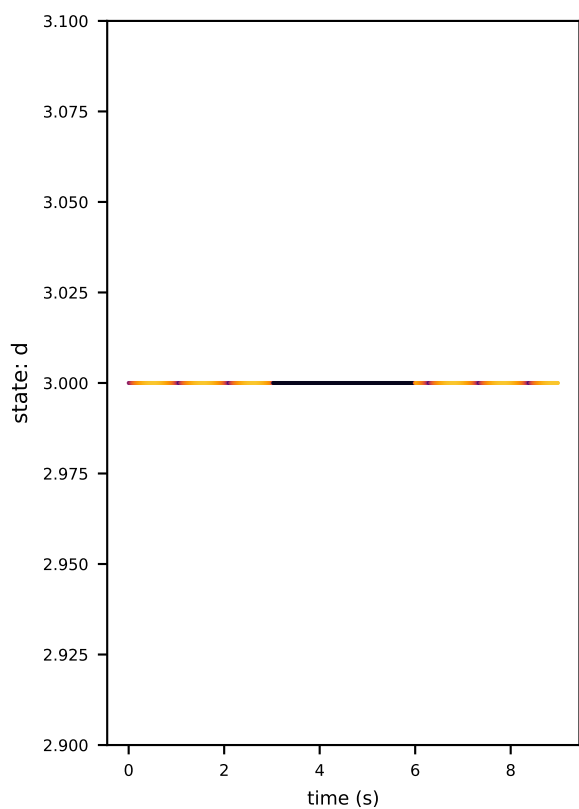
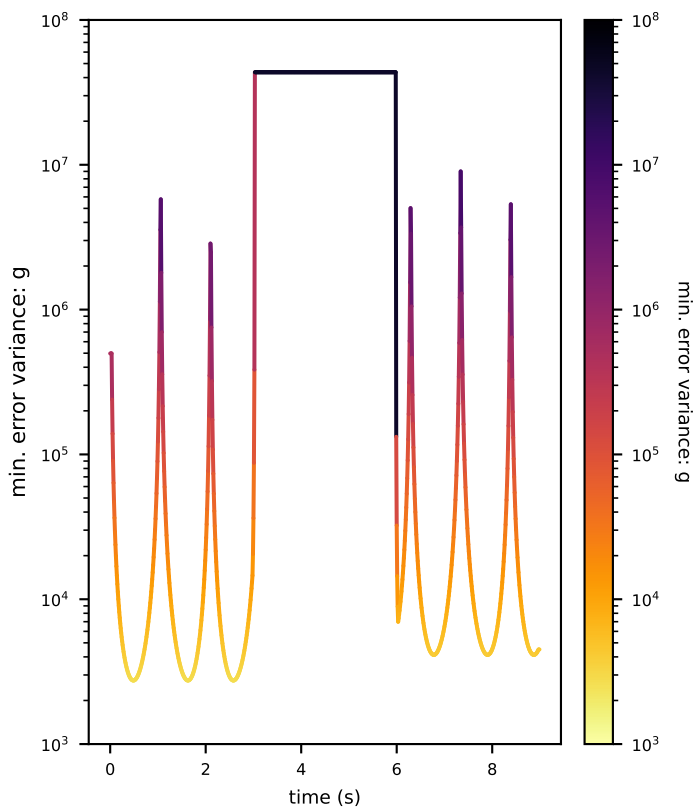
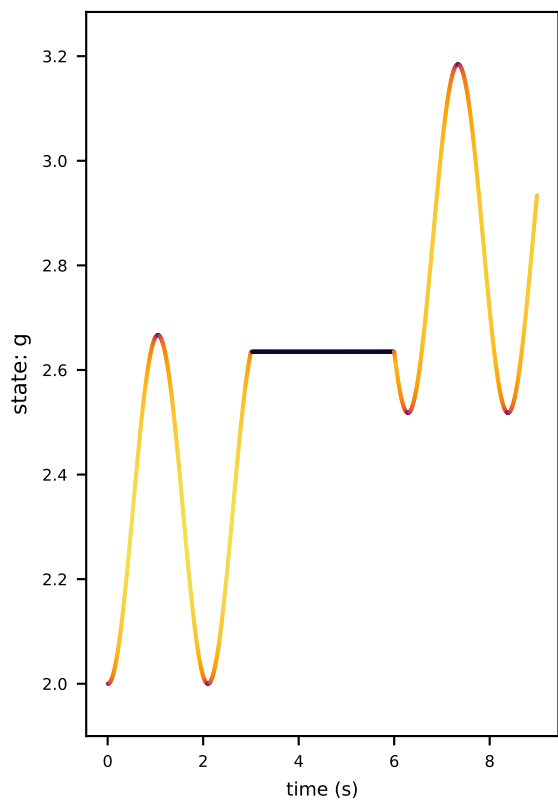
Pipeline Timing Summary

Trajectory: 900 time-steps | Sliding windows: 895 | Window size: 6



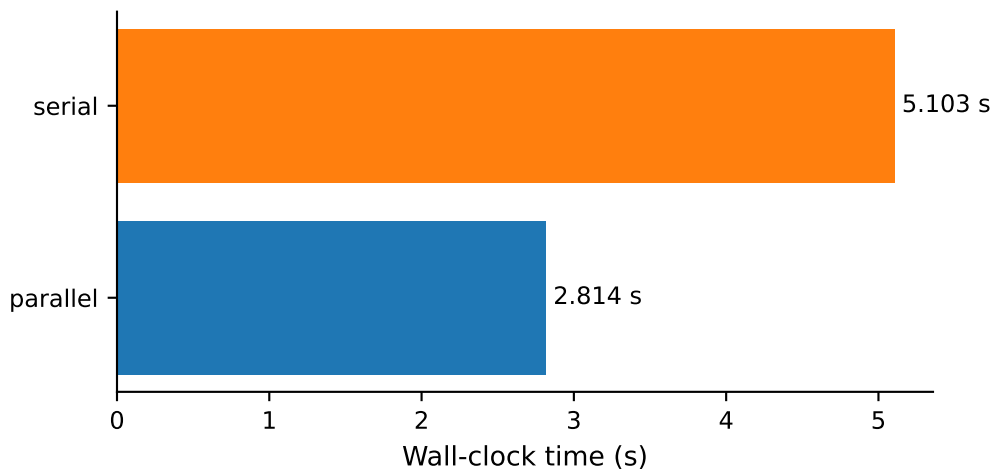
Step	Time (s)	% of total	Description
Simulator setup	0.007	0.1%	Construct Simulator: build do_mpc model, MPC controller
Simulation	0.183	3.0%	Open-loop simulate 900 steps (9.0 s of simulated time)
Sliding EOM	5.103	82.9%	SlidingEmpiricalObservabilityMatrix: 895 windows × 4 simulations each (2 perturbations × 2 states)
Sliding Fisher	0.427	6.9%	SlidingFisherObservability: $F = O^T R^{-1} O$ and $(F + \lambda I)^{-1}$ for 895 windows
Error variance	0.005	0.1%	get_minimum_error_variance() + NaN fill
PDF generation	0.427	6.9%	Render equations page, timing page, observability plot
TOTAL	6.153	100%	

Minimum Error Variance over Time (mono-camera system)



Parallel vs Serial Benchmark

SlidingEmpiricalObservabilityMatrix | 895 windows | parallel_sliding=True | simulator_factory=<callable> | n_workers=18 (multiprocessing.Pool, spawn)



Serial time: 5.103 s

Parallel time: 2.814 s

Speedup: 1.81×

Correctness: PASS — results match serial

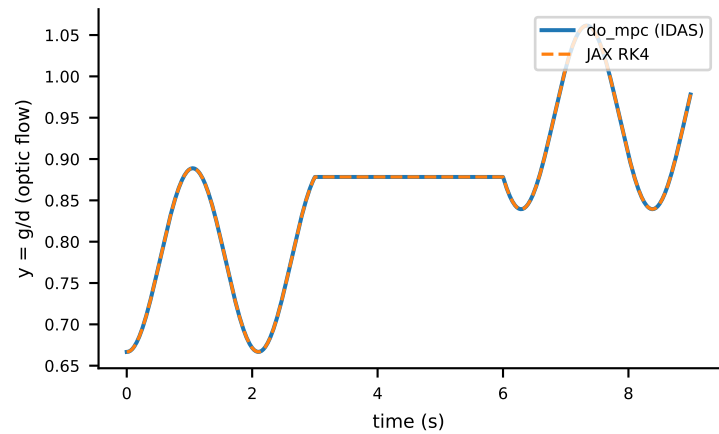
Notes

- Simulator.simulate() mutates shared state (self.simulator, self.x, ...) on each call.
CasADi/IDAS is NOT thread-safe: ThreadPoolExecutor causes fatal crashes.
- The new simulator_factory parameter passes a zero-arg callable to SEOM.
Each worker process calls factory() once to get its own Simulator.
This eliminates all shared state between workers.
- Uses multiprocessing.Pool with start method "spawn" (macOS/Windows default).
"spawn" avoids fork-safety issues with CasADi shared libraries.
- Each worker incurs ~0.02 s Simulator setup overhead (one-time, amortized across all windows assigned to that worker).

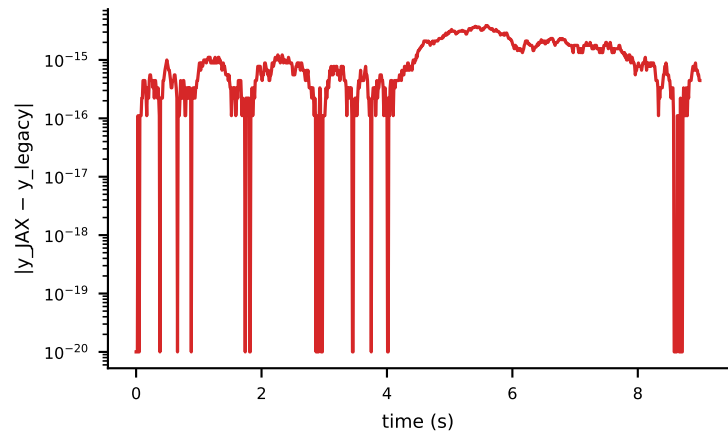
JAX Autodiff vs Legacy Numerical Observability

Mono-camera system | RK4 integrator | jacfwd Jacobian vs $\pm\epsilon$ perturbations

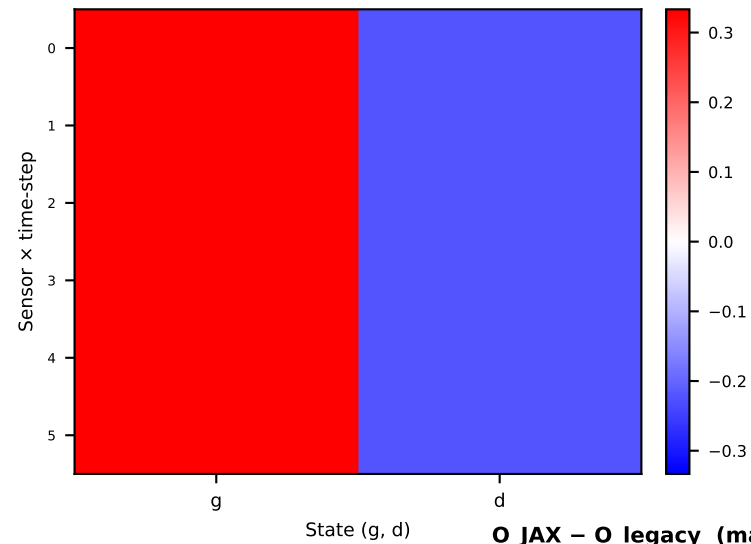
Simulation output: y trajectory



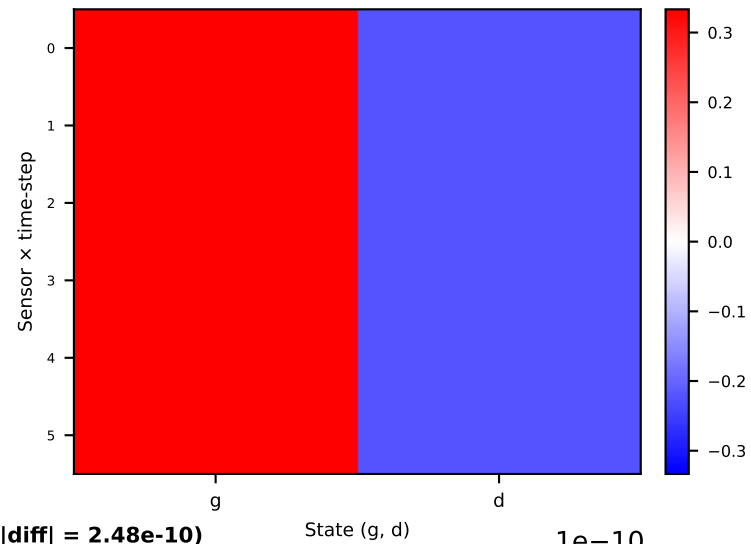
Residual (log scale)



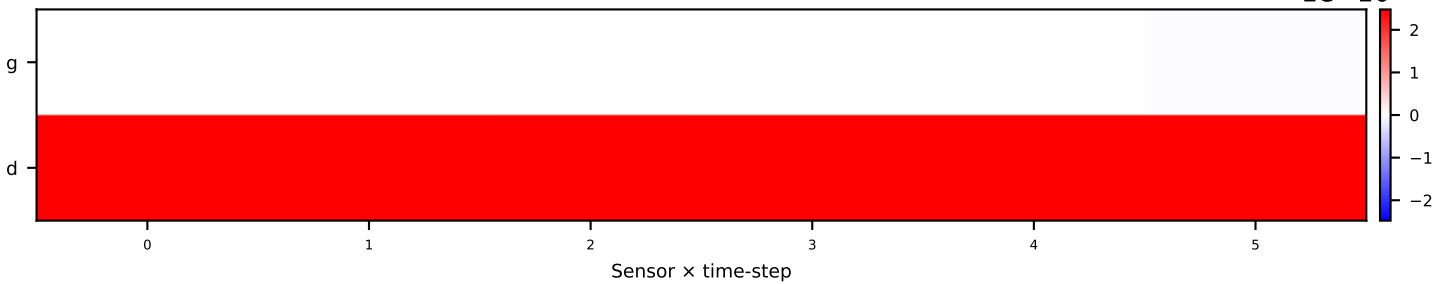
O legacy (numerical $\pm\epsilon$)



O JAX (autodiff jacfwd)



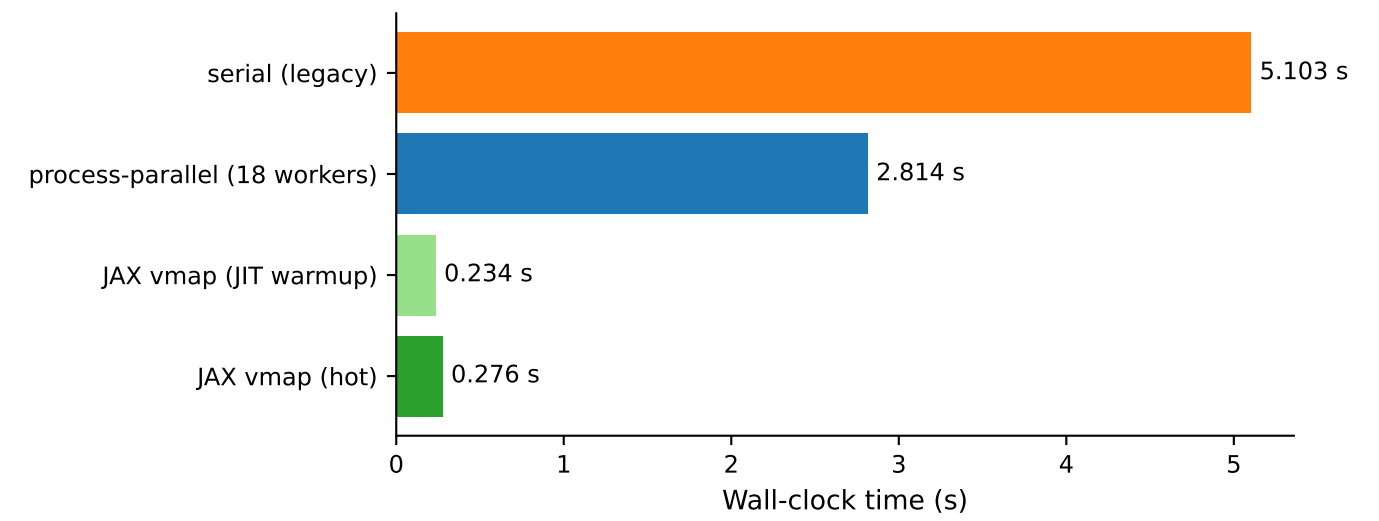
O_JAX - O_legacy (max |diff| = 2.48e-10)



Step	Time (s)	Speedup vs legacy	Notes
do_mpc simulate (full traj)	0.183	1.0× (baseline)	900 steps, IDAS solver
JAX RK4 simulate (JIT warmup)	0.055	3.3×	First call: traces + compiles XLA kernel
JAX RK4 simulate (hot)	0.0165	11.1×	Subsequent calls: runs compiled kernel
Legacy O, single window	0.0057	1.0× (baseline)	4 perturbation sims (2 states × $\pm\epsilon$)
JAX O = jacfwd (JIT warmup)	0.050	0.1×	First call: traces jacfwd + compiles
JAX O = jacfwd (hot)	0.0432	0.1×	Hot call; max O_diff = 2.48e-10

JAX vmap Sliding Observability Benchmark

SlidingEmpiricalObservabilityMatrix | 895 windows | window size: 6



Serial (legacy):	5.103 s	1.0× (baseline)
Process-parallel (18w):	2.814 s	1.8×
JAX vmap (JIT warmup):	0.234 s	21.8×
JAX vmap (hot):	0.2761 s	18.5×
Correctness:	PASS — max O_jax – O_serial = 3.95e-10	

How it works

- JaxSlidingEmpiricalObservabilityMatrix extracts all window initial states and input sequences into two batched arrays:

x0_batch

shape (n_windows, n)

– one x0 per window

u_batch

shape (n_windows, w, m)

– one input sequence per window

- A single vmapped+JIT-compiled jacfwd call computes all Jacobians at once:

vmapped_jac

= jax.jit(jax.vmap(jax.jacfwd(sim, argnums=0)))

jac_batch

= vmapped_jac(x0_batch, u_batch)

(n_windows, w, p, n)

- On CPU this becomes a single parallelisable XLA computation; on GPU it would map to a single kernel launch, giving further speedups.
- No process spawning, no Simulator cloning — pure JAX array operations.
- JIT warmup cost is paid once; all subsequent calls use the compiled kernel.